INCH-POUND

MIL-H-1411E
15 January 1991
SUPERSEDING
MIL-H-1411D
30 June 1986

MILITARY SPECIFICATION

HOOK, TENT, ASSEMBLY, WALL SECTION

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers a wall section tent hook assembly.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5017 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8340

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SPECIFICATIONS

FEDERAL

PPP-B-601 - Boxes, Wood, Cleated Plywood
PPP-B-636 - Boxes, Shipping, Fiberboard

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads
MIL-STD-731 - Quality of Wood Members for Containers and Pallets
MS 27040 - Nut, Plain, Square-Steel, Cadmium Plated
MS 35206 - Screw, Machine-Pan Head, Cross-Recessed, Carbon Steel, Cadmium Plated, UNC-2A (IN./MM)

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

U.S. ARMY NATICK RESEARCH, DEVELOPMENT, AND ENGINEERING CENTER

5-4-708 - Hook, Tent, Assembly, Wall Section; Style 1
5-4-709 - Hook, Tent, Assembly, Wall Section; Style 2

(Copies of drawings are available from the U.S. Army Natick Research, Development, and Engineering Center, ATTN: STRNC-UX, Natick, MA 01760-5017.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 148 - Steel Castings, High Strength, for Structural Purposes

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A 385 - Providing High-Quality Zinc Coatings (Hot-Dip)
D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

(Non-Government standards and other publications are normally available from organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3..

3.2 Materials. The materials used shall conform to the applicable referenced documents specified herein. It is encouraged that recycled material be used when practical as long as it meets the requirements of this specification.

3.2.1 Steel casting. Steel casting shall conform to grade 90-60 or 105-85 of ASTM A 148/A 148M.

3.2.2 Machine screw and nut. Machine screw and nut shall conform to MS 35206-231 and MS 27040-6 respectively.

3.3 Construction. Construction of the hook shall conform to either Drawing 5-4-708 or 5-4-709 and as specified herein. The hook shall be capable of withstanding a minimum pull of 150 pounds without fracture, permanent deformation, or separation when tested as specified in 4.4.4.

3.3.1 Hook body and clamp. The hook body and clamp shall be of cast steel as specified in 3.2.1.

3.4 Finish.

3.4.1 Castings. Castings for body and clamp shall be ground, filed, polished, or tumbled smooth and have a natural finish. Castings shall be zinc plated per ASTM A 385.

3.5 Workmanship. The completely finished tent hooks shall be free of blow-holes, porosity, hard spots, burrs, cracks, or sharp edges. Threads of screw and nut shall be free of damage to the extent that the nut cannot be tightened or loosened. Screw and nut shall be free from rust. The end item shall conform to the quality of product established by this specification and the occurance of defects shall not exceed the applicable acceptable quality levels.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for ensuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point, or at all points in the manufacturing process necessary to ensure compliance with all dimensional requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 3.1 and 6.2), it shall be examined for defects specified in 4.4.2 and 4.4.3 and shall be tested for the characteristic specified in 4.4.4. Any nonconformance shall be cause for the rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

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4.4.2 End item visual examination. The end items shall be examined for the defects listed in table I. The lot size shall be expressed in units of tent hooks. The sample unit shall be one tent hook. The inspection level shall be II, and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5 for major defects and 6.5 for total (major and minor combined) defects.

TABLE I. End item visual defects

Examine	Defect	Classification Major	Minor
Finish:			
Casting	Not zinc plated or partially zinc plated	101	
	Coating is blistered, pitted, not adherent, or showing evidence of corrosion	102	
Screw and nut	Not cadmium plated	103	
	Plating is blistered, pitted, not adherent, or not continuous	104	
	Evidence of corrosion	105	
Construction and workmanship	Not smooth		201
	Blow-holes, porosity, hard spots, burrs, flashings, cracks or sharp edges	106	
	Any component or part missing	107	
	Mating surfaces between clamp and body not flat	108	
	Not assembled as shown on applicable drawing		202
	Threads of screw or nut damaged to the extent that the nut cannot be tightened or loosened	109	
	End of screw not peened or insufficiently peened that could result in loss of nut		203
	Wrong type screw or nut		204

4.4.3 End item dimensional examination. The end items shall be examined for nonconformance to the dimensions specified on the drawings. Only those dimensions that can be evaluated without damaging or disassembling the end items shall be examined. Any dimension not within the specified tolerance shall be classified as a defect. The lot size shall be expressed in units of tent hooks. The sample unit shall be one tent hook. The inspection level shall be S-2, and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

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4.4.4 End item testing. Testing of the completely fabricated tent hook shall be performed in accordance with table II for the specified characteristic. The lot size shall be expressed in units of tent hooks. The sample unit shall be one tent hook. The inspection level shall be S-1. The failure of any pull test shall be cause for rejection of the lot.

TABLE II. End item test

Characteristic	Requirement paragraph	Test method	Number of determinations per sample unit	Results report as
Pull test load	3.3	4.5.1	1	Pass or fail

4.4.5 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application
Materials	Any component missing, damaged, or not as specified
Workmanship	Inadequate application of components, such as: incomplete sealing or closure of flap, improper taping, loose strapping or inadequate stapling Bulged or distorted container
Content	Number of hooks per unit pack and shipping container not as specified

4.4.6 Palletization examination. The fully packaged and palletized end items shall be examined for the defects listed below. The lot size shall be expressed in units of palletized unit loads. The sample unit shall be one palletized unit load, fully packaged. The inspection level shall be S-1, and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirement
Palletization	Pallet pattern not as specified Load not bonded as specified

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When it meets this requirement

Weight	Exceeds maximum load limits
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application

4.5 Method of inspection.

4.5.1 Pull test. The hook shall be subjected to the pull test load, as shown on Drawing 5-4-708 or 5-4-709, as applicable. The load rate shall not exceed 1000 pounds per minute until a load of 150 pounds is reached. After the load has been released, the hook shall be examined. Any hook having evidence of fracture, permanent distortion, or separation shall constitute failure of the test.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial, as specified (see 6.2).

5.1.1 Level A preservation. Unless otherwise specified (see 6.2), five hundred hooks of like description shall be unit packed in a close-fitting fiberboard box conforming to style RSC, type CF or SF, class domestic, grade 200 of PPP-B-636. Each box shall be agitated to ensure a compact and well-filled box. Closure shall be in accordance with the appendix of PPP-B-636.

5.1.2 Commercial preservation. Hooks shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or Commercial, as specified (see 6.2).

5.2.1 Level A packing. Unless otherwise specified (see 6.2), three thousand hooks of like description, preserved as specified in 5.1, shall be packed in a close-fitting cleated plywood box conforming to overseas type, style A or J, grade A or B, type 2 load of PPP-B-601. Each shipping container shall be completely filled, with no voids, and shall be of a size that can be palletized in accordance with MIL-STD-147. Shipping containers shall be closed, waterproofed, and reinforced in accordance with the appendix of PPP-B-601.

5.2.2 Level B packing. Unless otherwise specified (see 6.2), three thousand hooks of like description, preserved as specified in 5.1, shall be packed in a close-fitting fiberboard box conforming to style FTC, type CF, variety DW, or SF, grade 350 of PPP-B-636. Each shipping container shall be completely filled with no voids, and shall be of a size that can be palletized in accordance with MIL-STD-147. Each shipping container shall be closed and reinforced in accordance with the appendix of PPP-B-636.

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5.2.2.1 Weather-resistant shipping container. When specified (see 6.2), the shipping container shall be V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with the appendix of PPP-B-636.

5.2.3 Commercial packing. Hooks, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.3 Palletization. When specified (see 6.2), hooks packed as specified in 5.2, shall be palletized on a 4-way entry pallet in accordance with load type I or Ia (as applicable) of MIL-STD-147. Pallet type shall be type I (4-way entry), type IV or type V in accordance with MIL-STD-147. Pallets shall be fabricated from wood groups I, II, III, or IV of MIL-STD-731. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means C and D or film bonding means F or G. Pallet pattern shall be in accordance with the appendix of MIL-STD-147.

5.4 Marking. Unit packs, shipping containers and unit loads shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The tent hooks are intended to be used for hanging a tent wall from a tent roof.

6.2 **Acquisition requirements. Acquisition documents must specify the** following:

a. Title, number, and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. When a first article is required (see 3.1, 4.3, and 6.3).
d. Levels of preservation and packing (see 5.1 and 5.2).
e. Whether quantity per unit pack and pack is other than specified (see 5.1.1, 5.2.1, and 5.2.2).
f. When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).
g. When palletization is required (see 5.3).

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of Federal Acquisition Regulation (FAR) 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

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6.4 Subject term (key word) listing.

Attachment fixture

6.5 International standardization agreements. Certain provisions of this document are the subject of international standardization agreement as cited in NATO, STANAG NO. 2882, relative to camouflage requirements for tents, shelters, and subsidiary components. When amendment, revision, or cancellation of this document is proposed which will affect or violate the international agreement concerned, the preparing activity will take appropriate reconciliation action through international standardization channels including departmental standardization offices, if required.

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:

Army - GL
Air Force - 99

Review activities:

Army - MD
Air Force - 82
DLA - CT

Preparing activity:

Army - GL

(Project 8340-0542)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

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2. The submitter of this form must complete blocks 4, 5, 6, and 7.
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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-H-1411E	1991 January 15

3. DOCUMENT TITLE
HOOK, TENT, ASSEMBLY, WALL SECTION

4. NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.*)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
U.S. Army Natick RD&E Center	(1) Commercial 508-651-4531	(2) AUTOVON 256-4531
c. ADDRESS (Include Zip Code)	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:	

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